

JUNEAU, AK, USA

WMO: 703810

Lat: 58.357N

Lon: 134.564W

Elev: 5

StdP: 101.27

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25309

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-12.3	-23.7	0.4	-11.7	-20.6	0.6	-9.5	14.0	3.8	12.3	2.9	2.8	90	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	23.5	15.4	21.4	14.7	19.4	13.7	16.3	22.2	15.3	19.9	14.5	18.1	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.0	16.5	13.5	9.6	16.0	12.9	9.2	15.4	45.3	22.1	42.7	19.9	40.5	17.9	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.3	8.7	-17.3	26.8	2.5	1.8	-19.1	28.1	-20.6	29.2	-22.0	30.2	-23.8	31.5		
(5)			-17.8	17.7	2.4	0.9	-19.5	18.3	-20.9	18.9	-22.2	19.4	-24.0	20.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-1.8	-1.1	0.5	4.9	9.5	12.7	14.0	13.5	10.2	5.8	1.0	-1.0		
	DBStd	6.77	5.15	4.43	3.89	2.59	2.83	2.52	2.13	2.12	1.96	2.57	4.11	4.81		
	HDD10.0	1936	365	311	293	153	44	3	1	1	21	132	272	342		
	HDD18.3	4608	623	544	552	402	273	171	137	151	244	389	522	600		
	CDD10.0	371	0	0	0	2	28	83	123	108	26	1	0	0		
	CDD18.3	3	0	0	0	0	0	1	1	1	0	0	0	0		
	CDH23.3	66	0	0	0	0	2	26	25	13	0	0	0	0		
	CDH26.7	6	0	0	0	0	0	3	2	1	0	0	0	0		
	WSAvg	3.2	3.1	3.1	3.4	3.4	3.3	3.0	2.9	3.0	3.4	3.6	3.3	3.4		
	PrecAvg	1453	119	99	88	72	87	80	114	140	191	201	134	132		
(15) Precipitation	PrecMax	2163	231	215	162	190	234	153	263	313	385	387	281	346		
	PrecMin	1044	24	2	15	12	21	28	29	14	59	90	29	12		
	PrecStd	273	52	56	37	35	40	29	47	62	71	67	61	67		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	7.7	10.8	17.7	22.8	26.6	26.2	25.1	19.6	12.8	9.7	8.7		
		MCWB	6.3	4.7	6.9	9.6	12.7	15.9	16.6	16.4	12.5	10.4	8.0	7.0		
	2%	DB	6.4	6.2	8.0	14.0	20.2	22.9	23.2	22.6	16.9	11.2	8.0	7.0		
		MCWB	5.4	4.8	4.9	7.3	12.1	14.8	16.1	15.5	11.9	9.3	6.8	6.0		
	5%	DB	5.2	5.2	6.4	11.8	17.8	20.2	20.8	20.2	15.0	10.2	6.9	5.7		
		MCWB	4.4	4.1	4.3	6.7	10.8	13.9	15.2	14.9	11.8	8.6	5.8	4.9		
	10%	DB	4.1	4.1	5.3	9.8	15.5	18.0	18.4	17.9	13.6	9.4	5.8	4.5		
		MCWB	3.4	3.1	3.7	5.8	9.9	12.9	14.2	14.0	11.5	8.1	4.9	3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	5.9	7.2	10.4	14.0	16.6	17.6	17.4	14.5	11.3	8.2	7.6		
		MCDB	7.6	6.6	9.0	16.4	20.8	24.3	24.3	23.5	16.3	12.1	9.0	8.0		
	2%	WB	5.5	5.1	5.8	8.4	12.6	15.3	16.5	16.2	13.4	10.1	6.9	6.2		
		MCDB	6.2	5.9	7.4	12.7	18.7	21.8	22.5	21.2	15.1	10.8	7.8	6.9		
	5%	WB	4.4	4.2	4.9	7.4	11.5	14.3	15.3	15.3	12.5	9.1	6.1	5.0		
		MCDB	5.1	5.1	6.0	10.5	16.4	19.2	19.7	19.0	14.1	9.8	6.9	5.7		
	10%	WB	3.5	3.4	3.9	6.5	10.6	13.3	14.5	14.4	11.8	8.4	5.1	3.8		
		MCDB	4.1	4.2	5.0	9.1	14.5	17.2	17.8	17.2	13.1	9.2	5.7	4.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	5.4	6.1	7.9	8.6	7.7	6.8	6.8	5.8	4.9	4.4	4.2		
		MCDBR	4.0	5.5	7.9	12.6	14.4	13.5	13.1	12.8	10.0	6.1	4.2	4.0		
		MCWBR	3.2	4.1	5.3	7.0	7.1	6.6	6.8	6.9	5.9	4.3	3.2	3.3		
		MCDBR	3.8	4.6	6.3	10.1	11.9	11.8	10.9	10.4	7.1	4.5	3.7	3.9		
	5% WB	MCWBR	3.2	3.7	4.7	6.0	6.1	6.0	6.1	6.1	4.7	3.7	3.2	3.3		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.353	0.329	0.394	0.412	0.425	0.429	0.390	0.374	0.330	0.287	0.266		
	taud		2.079	1.997	2.155	2.000	2.072	2.151	2.198	2.347	2.322	2.310	2.127	2.045		
	Ebn at Noon		506	623	803	799	814	811	797	807	754	678	509	407		
	Edh at Noon		54	95	106	147	147	138	129	104	91	68	50	39		
(44) All-Sky Solar Radiation	RadAvg		0.38	1.10	2.31	3.86	4.93	4.93	4.21	3.45	2.19	1.15	0.51	0.26		
	RadStd		0.05	0.20	0.32	0.45	0.55	0.37	0.47	0.44	0.36	0.18	0.08	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page